

Day 1 Problem Set

Geometry Summer Credit | Coordinate Geometry and Angle Relationships

Name:	Date:	Period:
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Goal

Practice distance, midpoint, angle relationships, and equation setup. Show work clearly enough that someone else can follow your reasoning.

Support Practice: Build the Pieces

1.	Compute $ 8 - 3 $, $ 3 - 8 $, $ -2 - 5 $, and $ 5 - (-2) $.
2.	Simplify $\sqrt{36}$, $\sqrt{45}$, $\sqrt{75}$, and $\sqrt{98}$.
3.	Solve $x + 42 = 90$.
4.	Solve $4x - 9 = 83$.

Core Practice: Coordinate Geometry

5.	Find the distance between $(4, 7)$ and $(4, -2)$.
6.	Find the distance between $(-6, 3)$ and $(5, 3)$.
7.	Find the distance between $(0, 0)$ and $(8, 15)$.

8. Find the distance between $(-3, -4)$ and $(5, 2)$. Leave the answer exact.

9. Find the midpoint of $(10, -6)$ and $(-2, 8)$.

10. Find the midpoint of $(-7, -1)$ and $(9, 5)$.

11. Point $M(2, 3)$ is the midpoint of AB . If $A(-4, 9)$, find B .

Day 1 Problem Set

Continued

Core Practice: Angle Relationships

12. Angles A and B are complementary. If $m\angle A = 34$ degrees, find $m\angle B$.

13. Angles C and D are supplementary. If $m\angle C = 118$ degrees, find $m\angle D$.

14. Two angles form a linear pair. One angle is $7x + 8$ and the other is $3x + 12$. Find x and both angle measures.

15. Two vertical angles are labeled $5x + 11$ and $8x - 22$. Find x and the angle measure.

16. Two complementary angles are labeled $2x + 5$ and $3x + 10$. Find x and both angle measures.

17. Error analysis: A student says vertical angles always add to 180 degrees. Explain the mistake and give a correct statement.

Mixed Review and Reasoning

18. The endpoints of a diameter of a circle are $(-4, 1)$ and $(6, 9)$. Find the center of the circle.

19. A segment has endpoints $P(-2, 5)$ and $Q(10, -4)$. A student says the midpoint is $(8, 1)$. Identify the error and correct it.

20. Create your own linear-pair problem with algebra. Then solve it.

21. Challenge: Points A(1, 2), B(7, 10), and C(13, 18) are plotted. What do you notice about the distances AB and BC? What might that suggest about the three points?

Before You Submit

Circle one problem you feel confident about and star one problem you want help with.